Bruno COSTACÈQUE

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Education

2021 – 2024 ***PhD in Mathematics***, *Télécom Paris*, Palaiseau.
Steins method for extreme value distributions, under the supervision of Laurent DECREUSEFOND

2020 – 2021 ***Master 2 Probabilités et Finance (El Karoui)***, *Université Pierre et Marie Curie, École Polytechnique*, Paris, Palaiseau.
Courses: derivatives, Monte Carlo methods, high frequency trading, stochastic calculus, stochastic control, risk measures, extreme value distributions, machine learning, jump processes, Malliavin calculus, mean field games, optimal stopping, longevity risk

2019 – 2020 ***Agrégation de mathématiques – Option probability and statistics***, *École Normale Supérieure Paris-Saclay*, Cachan.
Intensive preparation in graduate level Mathematics (algebra, analysis, probability, statistics, programming, *etc.*) for competitive entrance examination to become a teacher (*professeur agrégé*) – passed (ranked 162/307)

2018 – 2019 ***Master Hadamard*, first year**, *École Normale Supérieure Paris-Saclay, École Polytechnique, Paris-Sud (Paris XI)*, Cachan, Palaiseau, Orsay.
Courses: abstract algebra (characters, modules, Galois theory), probability theory (martingales, Markov chains), functional analysis, harmonic analysis, spectral theory, PDE, statistics, random graphs

2017 – 2018 ***Bachelor of mathematics***, *École Normale Supérieure Paris-Saclay, Paris-Diderot (Paris VII)*, Cachan, Paris.
Courses: abstract algebra (groups, rings, fields), measure theory and probability, numerical analysis (for ODE and PDE), Fourier analysis, quantum mechanics, complex analysis, advanced probability theory (concentration inequalities, random matrices)

2016 – 2017 ***Bachelor of economics***, *École Normale Supérieure Paris-Saclay, Paris-Sorbonne (Paris I)*, Cachan, Paris.
Courses: macroeconomics, microeconomics, game theory, finance, corporate finance, operational research, management, marketing, accounting

2013 – 2016 ***Preparatory Classes – ECE***, *Lycée Saint-Jean*, Douai.
Intensive undergraduate level preparation in Mathematics and Economics for competitive entrance examinations to top French Business Schools. Admitted to ENS Paris-Saclay and ranked 15th.

Experience

April 2025 – August 2028 **Research fellow**, *School of Physical and Mathematical Sciences, NTU*, Singapore.
A deep branching solver for fully nonlinear partial differential equations PI: Nicolas PRIVAULT

May 2024 **Interviewer**, *SKEMA Paris*, Paris.
Interviewer for the oral examination to enter Skema Business School

April – September 2021 **Research intern**, *Natixis*, Paris.
Differential machine learning and xVA, under the supervision of Rida EL KARAFLI

April – July 2019 **Research intern**, *Department of Statistics*, Oxford.
Branching random walks with selection, under the supervision of Julien BERESTYCKI

January – June 2018 **Research intern**, *Centre de mathématiques et de leurs applications (CMLA)*, Cachan.
Monte Carlo methods and non-reversible Markov chains, under the supervision of Alain DURMUS

September 2016 – August 2021 **Civil servant**, *École Normale Supérieure Paris-Saclay (ENS PS)*, Cachan.
Élève normalien

Teaching experience

August 2025 – Present **Teaching assistant**, *NTU*, Singapore.
Subject: Real analysis (Chek Beng CHUA)

October 2021 – December 2024 **Teaching assistant**, *Télécom Paris*, Palaiseau.
Subjects: Probability (Pascal BIANCHI), asymptotic statistics (Pascal BIANCHI), convex optimization (Olivier FERCOQ), high dimensional probability (Yann ISSARTEL), numerical analysis and Monte Carlo methods (Olivier FERCOQ), statistics (Philippe CIBLAT)

Talks and conferences

April 2026 **CREAR WG Risk**, *ESSEC Asia Pacific*, Singapore.
Topic: Stein's method for extreme value analysis

December 2025 **SWGEN 2025 workshop**, *Institute of Environmental Geosciences*, Grenoble.
Topic: Pareto versus Gaussian: rates of convergence for large sums of regularly varying random variables

December 2024 **Paris-Saclay seminar for PhD students in mathematics**, *Paris-Saclay University*, Orsay.
Topic: The generator approach of Stein's method and the extreme value theorem

October 2024 **SAFE Workshop**, *NTU-CNRS*, Singapore.

September 2024 **SAMA at Aussois**, *LSCE*, Aussois.
Topic: The generator approach of Stein's method and the extreme value theorem

June – July 2022 **Steins Method: The Golden Anniversary**, *NUS*, Singapore.

October 2021 **Journée de rentrée à l'IHES**, *IHES*, Bures-sur-Yvette.

Publications

August 2026 **B. Costacèque, N. Privault**, *Convergence rates for the extreme value theorem via Stein's method*, Submitted to Extremes.
`https://arxiv.org/abs/2608.17562`

November 2025 **B. Costacèque, L. Decreusefond**, *Convergence rate for the coupon collector's problem with Stein's method*, Stochastic Processes and their Applications.
`https://arxiv.org/abs/2501.06535`

September 2025 **B. Costacèque, L. Decreusefond**, *Functional analysis of multivariate max-stable distributions*, arXiv preprint.
`https://arxiv.org/abs/2509.02200`

July 2025 **B. Costacèque, L. Decreusefond**, *Stein's method for max-stable random vectors*, arXiv preprint.
`https://arxiv.org/abs/2507.00463`

Computer skills

Basic Scilab, C++, VBA
Intermediate R
Advanced Python, LaTeX

Languages

French Native speaker
English Advanced C1 Level (Cambridge Assessment English)
German Intermediate B2 Level (CEFR)
Japanese Elementary (JLPT N4)

Certifications

Machine learning	Coursera, Andrew NG, Certificate No. 83K98WJB6U67
Japanese	Japanese Language Proficiency Test, N4
English	Cambridge Assessment English, C1

Interests

- Chess
- Swimming
- Go
- Drawing